

SEVEN SEGMENT DISPLAY

Major Project

Design an Arduino-based embedded program using a single-digit seven-segment display. The program should sequentially turn ON each segment one by one, starting with all segments OFF. Implement a fixed, predefined delay between each segment activation. Once all seven segments are ON, reverse the pattern by turning OFF each segment sequentially until all segments are OFF again. The sequence should then repeat continuously.

SYSTEM OVERVIEW

The system is designed using an Arduino UNO board connected to a common anode 7-segment display. Each segment of the display is connected to different digital pins of the Arduino.

In a common anode configuration:

- Logic HIGH (5V) → Segment OFF
- Logic LOW (0V) → Segment ON

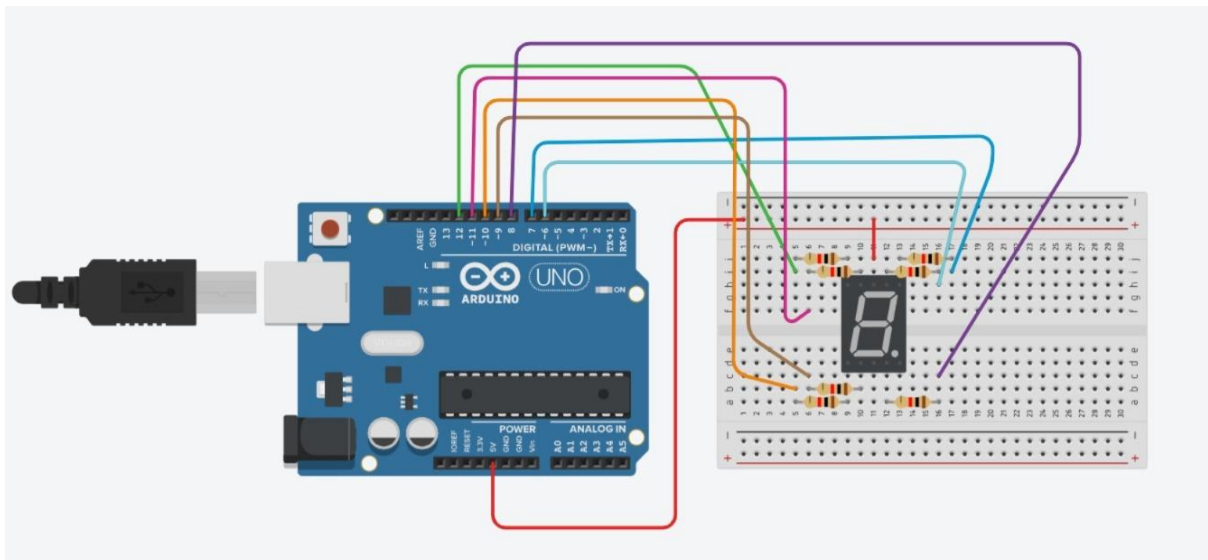
COMPONENTS USED

- Arduino UNO
- Single digit 7-segment display (Common Anode)
- 220Ω / 330Ω resistors (current limiting)
- Breadboard
- Jumper wires
- USB cable for Arduino power/programming

HARDWARE DESCRIPTION

Arduino UNO	Main Controller
7 – Segment Display	Output Device
Current limiting Resistors	Protect LEDs from excess current
Jumper wires & Breadboard	Connections

CIRCUIT IMAGE



INITIAL STATE

At the beginning of the program, all segments of the 7-segment display are set to OFF state. Since it is a common anode display, all segments are initialized to HIGH condition to ensure no segment is glowing when the system starts.

WORKING PRINCIPLE

The working of the system is based on sequential control logic:

1. Initially, all segments are set to OFF state.
2. The Arduino activates each segment one by one from a to g, creating a forward sequence.
3. Each segment remains ON for a fixed delay period to create a visible effect.
4. After completing the ON sequence, the system waits for a short delay.
5. Then, the segments are turned OFF in reverse order from g to a.
6. This cycle repeats continuously.

ALGORITHM

- Initialize all segment pins as OUTPUT
- Set all segments to OFF state
- Turn ON segments sequentially (a $\rightarrow$ g)
- Introduce delay between each step
- Wait for a short duration
- Turn OFF segments in reverse order (g $\rightarrow$ a)
- Repeat continuously in loop

PROGRAM

```
// Arduino 7 Segment Display
```

```
// Sequential Segment ON/OFF
```

```
// Common Anode
```

```
int segPins[] = {6,7,8,9,10,11,12};
```

```
// a,b,c,d,e,f,g
```

```
int delayTime = 500;
```

```
void setup()
```

```
{
```

```
  for(int i = 0; i < 7; i++)
```

```
  {
```

```
    pinMode(segPins[i], OUTPUT);
```

```
    // Initially OFF
```

```
    digitalWrite(segPins[i], HIGH);
```

```
  }
```

```
}
```

```
void loop()
```

```
{
```

```
  // ON sequence
```

```
  for(int i = 0; i < 7; i++)
```

```
  {
```

```
    digitalWrite(segPins[i], LOW);
```

```
    delay(delayTime);
```

```
  }
```

```
  delay(1000);
```

```
  // OFF sequence (reverse order)
```

```
for(int i = 6; i >= 0; i--)  
{  
    digitalWrite(segPins[i], HIGH);  
    delay(delayTime);  
}  
delay(1000);  
}
```

DELAY INFORMATION

A fixed delay of 500 milliseconds is used between each segment activation and deactivation. This delay provides a visible sequential lighting effect and ensures proper observation of each segment transition.

Hint : This program demonstrates sequential control of a single-digit seven-segment display using a single Arduino board. The logic is based on digital pin communication to achieve ON/OFF sequencing of segments.

Simulation Link :

<https://www.tinkercad.com/things/d8Ra5WlePoT-arduino-7-segment-display/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fdesigns%2Fall>